Humans Are More Diverse: Audited Frontier LLMs Show Extreme Policies in Idealised AI Development Races

Anonymous Author(s)
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ABSTRACT

An AI development race creates a multi-agent safety dilemma. Each company can develop slowly and safely, or move faster while taking a risk that may remove its final reward. We use this repeated game to study strategic safety behaviour among large language model (LLM) agents in races with two to five players. However, a valid action does not show that an agent understands the game. We therefore place an audit gate before behavioural interpretation. We first verify the game engine, then test rule recall, state tracking, payoff calculation, and stability under different but equivalent task descriptions. We then compare LLM action sequences with an evolutionary game-theory benchmark and published human data, and explore differences across models, risk conditions, personas, and two- to five-player races. The multiplayer comparison is an exploratory supplementary pilot because the position analysis covers only two checkpoints. We audit nine frontier routes on the same frozen probe bank. All nine read the payoff matrix perfectly and eight of nine recall the rules, yet accuracy at rebuilding the current state ranges from 100% down to 20%, and no route answers three quarters of the closed-form expected-payoff probes. Five routes pass the gate. The action format, by contrast, was satisfied every time, so a well-formed action carries no information about whether the agent tracked the state behind it. That distinction changes what the behaviour means. Measured directly on action sequences, human participants are more diverse than every checkpoint that passes the gate, while the one checkpoint whose diversity matches the human sample is also the one that fails the gate most severely. Endpoint probe accuracy also ranks with how sharply play responds to the assigned risk, though the two groups overlap. A selection-strength sweep further shows that whether these agents appear to contradict the evolutionary benchmark depends on the regime it is read in. Multi-agent AI-race simulations therefore need validity checks and trajectory-level analysis before their outputs are called strategic, human-like, or safety-aware. Our findings apply only to the tested routes, prompts, and decoding settings.

KEYWORDS

Large language models, AI development races, multi-agent systems, repeated games, AI safety, task validity

1 INTRODUCTION

Competition over advanced artificial intelligence can create a safety dilemma. Each company may benefit from moving faster, but safety work can slow it down when its rivals do not follow the same standard [5, 6, 10]. An *AI development race* is a simple game that represents this tension. In each round, every player chooses between slower Safe development and faster Unsafe development [11, 16]. The game is idealised: it does not model a real AI company in full. Its value is control. It lets us test how choices change with competition, relative progress, and the earlier actions of other players.

Human experiments show why the order of actions matters. In the two-player study of Domingos and Han [14], the preregistered comparison between the two higher private-risk conditions found no clear difference. Exploratory analyses instead linked later Unsafe choices to three parts of the race history: the opponent's previous action, whether the player was ahead or behind, and the player's first action. We call this *path dependence*: an early choice or event changes the later path of the interaction. A useful human and LLM comparison must therefore examine full action sequences, which we call *trajectories*, rather than only the overall Unsafe rate.

Large language models (LLMs) can act as language-based players in this game. They are already studied as simulated participants, decision-making agents, and game-playing systems [1, 2, 4]. Yet their use creates a measurement problem. A model may return a correctly formatted action without correctly tracking the state or calculating the payoff. In a repeated game, one such action changes the next state and can affect every later round. A plausible final trajectory is therefore not, by itself, evidence that the model understood the game.

Prompt-based evaluation is itself sensitive to presentation choices that do not change what the task asks. Meaning-preserving formatting changes produce large, model-dependent spreads [25]; option order, response-token identity, output-format requests and even whitespace shift predictions [22, 24, 27]; and a persona prompt is an intervention rather than an improvement, its average benefit being absent or slightly negative [31]. The same threat appears in strategic choice specifically. Reversing action labels or reassigning their payoffs changes choice frequencies in Stag Hunt and Prisoner's Dilemma presentations [17], decisions move with topic, actor and world framing across vignettes built on one underlying game [23], and performance falls consistently when a familiar default mapping is replaced by an explicitly stated alternative [28]. This is why we audit wording, answer order, arithmetic disclosure, narrative framing and opaque response codes separately, and why we do not treat one familiar presentation as a neutral measuring instrument.

Reproducible game-based frameworks approach the problem from the other side. FAIRGAME standardises repeated-game experiments across models, languages and persona conditions and compares them with game-theoretic predictions [8]; later work adds a payoff-scaled Prisoner's Dilemma, a three-player Public Goods Game, and supervised recognition of canonical strategies from action trajectories [18, 19], and finds that cooperation can move in the opposite direction from its evolutionary benchmark. That motivates our use of full trajectories and theoretical benchmarks. Our audit asks the prior question those studies do not: whether the agent can track the race that produced the trajectory at all.

This distinction is easy to lose. An agent can produce an aggregate UNSAFE rate close to the human one while relying on incorrect state computations, prompt-specific heuristics, or unstable action encodings, and two checkpoints with nearly equal rates can respond quite differently to an opponent, to relative position, or to an early event. Aggregate similarity is therefore not evidence that an agent understands the task, reproduces human decision dynamics, or holds a general safety preference [7].

We address this problem with an *audit-first evaluation*: we check that the game and the agent's use of its rules are valid before we interpret the agent's behaviour. We reproduce the two-player game studied by Domingos and Han [14] and add a compatible multiplayer version based on Han et al. [16]. All agents in a round see the same state before anyone acts. Their choices remain hidden until every agent has responded, so the choices are simultaneous. The game engine, not the LLM, then calculates progress, payoffs, the end of the race, prizes, ties, and setbacks. We save every prompt, response, parsed action, retry, state change, configuration, random seed, and final outcome.

The audit has four levels. First, *mechanical validity* means that the game engine applies the stated rules correctly. Second, *task validity* means that an agent can recall the rules, rebuild the current state, apply a state change, and calculate the final result. Third, *representation robustness* means that choices remain stable when wording or response labels change but the game itself does not. Fourth, we compare valid trajectories with published human patterns. The human data are a reference for observable behaviour. They are not a target that an LLM must copy, and LLM outputs are not treated as replacements for human participants.

Two lessons from LLM-based behavioural simulation shape that boundary. The first is that a plausible output is not a comprehended task: qualitative findings have been reproduced in three of four experiments while the fourth showed a hyper-accuracy distortion [1], and a model can predict an opponent's pattern without acting on its own prediction [2]. Neither a plausible trajectory nor a verbal strategy report is therefore treated here as evidence of comprehension.

The second is that matching a mean is much weaker than matching a distribution. Market simulations recover the broad contrast between feedback conditions while showing less strategy heterogeneity than human participants [13], and the same gap appears across cognitive phenotyping [12], large survey populations [9, 26], and persona-based simulation of human perception [15], which is why position papers frame social simulation as defensible for exploratory use until validated against the population of interest [3]. Strategic-game evidence is similarly unsettled: repeated Prisoner's Dilemma policies shift with parameters and framing [21], human-like heuristics can be applied more rigidly than people apply them [30], agents may cooperate instead of reaching the predicted equilibrium [29], and multi-turn action sequences remain hard to reproduce from prompts alone [20]. Our human comparison therefore reports within-population diversity, not only aggregate UNSAFE rates.

We ask two questions. First, can an agent track the game well enough for its choices to be interpreted strategically at all, and are those choices stable across presentations that leave the game unchanged? Second, given that answer, how do agents respond to risk, to opponent history, and to relative progress, compared with the game-theory and human benchmarks?

The order matters, because the audit changes what the behaviour means. The tested routes express no common strategic policy: routes with similar aggregate UNSAFE rates respond differently to risk, to opponents, and to position, and rule recall coexists with incorrect state tracking. Measured directly on action sequences, human participants are more diverse than every route that passes the audit, while the one route whose diversity matches the human sample is the one that fails the audit most severely. Probe accuracy also ranks with how sharply a route responds to risk. Our contribution is the combination: an auditable race mechanism, a validity gate applied before any strategic reading, and a comparison against game-theory and human benchmarks made only for the routes that pass it. Exploratory multiplayer results remain in the supplementary material, because the available position analysis covers two checkpoints and cannot identify a general group-size effect.

Each reported behavioural conclusion is scoped to its tested checkpoint, prompt version, decoding contract, experimental configuration, and run period. The labels SAFE and UNSAFE denote actions within the experimental game and should not be interpreted as direct measurements of general LLM safety, subjective risk preference, or suitability for autonomous deployment.

2 PRELIMINARIES

We first specify the two-player game reproduced from Domingos and Han [14] (§2.1), then its generalisation to $N > 2$ companies (§2.2). Every quantity that is not restated in §2.2, namely the horizon, prize split, and private-risk formula, carries over unchanged from the two-player game; only the stage payoff changes shape (Figure 1).

2.1 Two-player game

Two companies $i \in \{1, 2\}$ play a repeated race. In every round t, both companies simultaneously choose an action $a_i^t \in \{S, U\}$ (Safe, Unsafe) and commit to it before either learns the other's choice for that round; only once both actions are committed does the round resolve (Figure 1a). Resolution has two parts. First, Safe advances the company's cumulative progress by $\sigma(S) = 1.0$ and Unsafe advances it by $\sigma(U) = 1.5$, so Unsafe always closes the race faster; progress accumulates as $P_i^t = P_i^{t-1} + \sigma(a_i^t)$. Second, the round pays a stage payoff $\pi_i^t = \pi(a_i^t, a_{-i}^t)$, read off the fixed matrix (own action by row, opponent action by column, Figure 1b):

$$\pi = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}. \tag{1}$$

Unsafe strictly dominates Safe at the stage-game level (row 2 exceeds row 1 in both columns): the only cost of Unsafe development enters later, through accumulated private risk rather than through the round payoff itself.

The race lasts a minimum of 5 rounds; after every completed round from round 5 onward it terminates with probability 0.2, giving an expected horizon $\mathbb{E}[T] = 9$. Neither company is ever told the horizon in advance, so a company can never condition its action on knowing the current round is the last one (Figure 1a). Termination then resolves in two further steps. First, the company with the higher cumulative progress P_i^T is declared the winner of

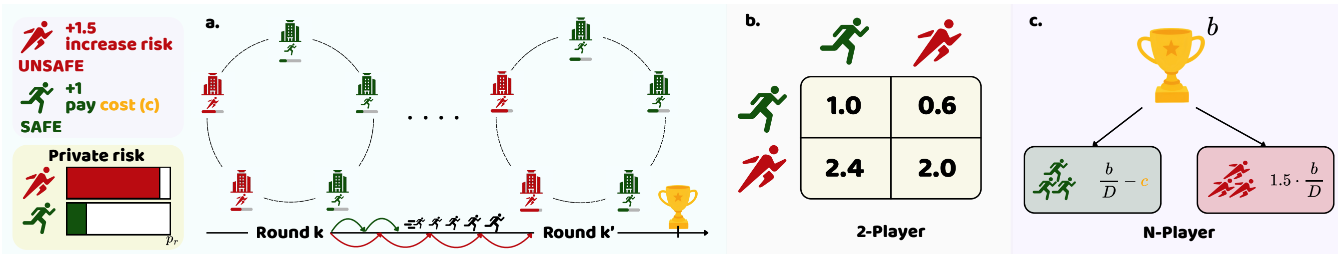

Figure 1: The AI-race mechanism, shared by the two-player and N-player games. (a) Each round every company advances by choosing Safe (green, slower, pays a cost) or Unsafe (red, faster, adding private setback risk); actions are simultaneous, the stopping time is hidden, and terminal prize B goes to the progress leader or is divided among tied leaders. (b) The two-player stage-payoff matrix, Equation 1. (c) The N-player stage-payoff rule, stated in full in the supplementary material, in which the per-round benefit b is distinct from terminal prize B.

a prize $B = 100$; if progress is exactly tied, the two companies split it 50/50. Second, and only for that winner or tied winners, an accumulated private setback risk is drawn:

$$q_i(T) = p_r^{\max} \cdot \frac{n_i^U(T)}{T}, q_i(T) \in [0, 1], p_r^{\max} \in \{0.10, 0.60, 0.90\}, \quad (2)$$

where $n_i^U(T)$ counts company i's Unsafe choices over the whole race and $p_r^{\max}$ is a treatment-level constant, assigned once per race, that we vary across three risk levels. If the draw against $q_i(T)$ fails, that company's entire race payoff, every stage payoff it accumulated plus the prize, is zeroed; if it succeeds, the company keeps both. A company that ends behind is never subject to this draw at all: it keeps its accumulated stage payoffs regardless of how much Unsafe it played, but receives no share of the prize (Figure 1a). This asymmetry, in which risk is built by both companies but realised only against whoever is ahead, is what makes Unsafe development a genuine gamble rather than a free speed advantage.

2.2 N-player game

The N-player game changes exactly one thing. Simultaneous sealed actions, the progress increments, the hidden minimum-five-round horizon, the terminal prize $B = 100$, and the risk formula of Equation 2 all carry over unchanged; the prize is split evenly among however many companies share the highest progress. What cannot carry over is the stage-payoff matrix, because with more than two companies there is no single opponent to index it against. Equation 1 therefore generalises to the group-count rule of Han et al. [16], in which both payoffs depend on how many of the N companies chose Safe in that round and not on which ones did (Figure 1c). The engine supports $N \in \{2, 3, 4, 5\}$. Since the multiplayer runs are an exploratory extension, the rule, its parameters, and its results are given in full in the supplementary material, and the main paper reports the two-player game that reproduces the human-study mechanism.

2.3 Game-theoretic benchmark

The game contains two opposing incentives. In a single round, Unsafe strictly dominates Safe: it gives more progress and a higher stage payoff against either action. Across the full race, however, Unsafe choices increase the setback risk for a winner or tied winner. The best action can therefore depend on the risk condition, earlier actions, and the expected response of other players. This is the strategic tension that the LLM agents must navigate.

We use four simple strategies as a theory benchmark: Always Safe (AS), Always Unsafe (AU), Conditional Safe (CS), which starts Safe and then copies the opponent, and Conditional Unsafe (CAS), which starts Unsafe and then copies the opponent. In the reconstructed finite-population evolutionary model of Domingos and Han [14], the most common strategy changes from AU at $p_r^{\max} = 0.10$, to CAS at 0.60, and to CS at 0.90. Thus, the theory does not predict one fixed amount of Unsafe play. It predicts a risk-dependent shift between unconditional speed, an aggressive opening, and conditional restraint. We use this as a qualitative benchmark, not as a fitted model of LLM output: prompted self-play trajectories are not samples from an evolutionary population.

3 EXPERIMENTAL DESIGN AND EVIDENCE POLICY

3.1 Agent protocol and units of analysis

At each decision, an agent receives a versioned prompt containing the current round, assigned maximum private risk, the publicly disclosed state, and the preceding revealed action profile when $t > 1$. Same-round decisions are sealed: every response is obtained from the same pre-action snapshot before the engine resolves actions, progress, payoffs, and risk. The environment, rather than generated text or arithmetic, determines all transitions. Raw response, parsed action, retry count, parser status, pre-turn state, terminal record, configuration, and seed allocation are retained for every decision.

The main behavioural outcome is whether an agent chooses Unsafe. The race, not the individual decision, is the main independent experimental unit. Decisions from the same race depend on earlier states and therefore cannot be treated as separate random samples. We report the number of races, trajectories, and decisions. Where possible, uncertainty is grouped by race or matched repetition. A *parse failure* occurs when a response does not follow the required action format. Because the resulting fallback action changes later states, one parse failure marks the whole race as contaminated.

Table 1: **Evidence strata used in this paper. Pilot modules are never pooled with confirmatory inference.**

Module	Primary units	Observed outputs	Use
Frontier endpoint admission (two admitted routes)	120 probe responses	120 answers	confirmatory gate
Calculator diagnostic	60 races	1,116 decisions	paired diagnostic
Context pilot (temperature 0)	768 races	13,680 decisions	robustness diagnostic
Fixed-state replay	1,536 state cells	1,536 responses	direct prompt diagnostic
Five-checkpoint baseline	150 races	2,790 decisions	descriptive only
OpenAI $N = 3$ to 5 baseline	180 races	6,120 decisions	exploratory only

3.2 Evidence strata and admission

We separate mechanical validity, task validity, diagnostic pilots, and confirmatory evidence (Table 1). A *diagnostic pilot* is a small exploratory test for possible effects or failures; it does not estimate a stable effect. A *confirmatory* study tests a question under a plan fixed before any result is inspected, and a run enters that stratum only when the protocol, route identifier, prompt and configuration hashes, race count, and exclusion rules were all fixed and recorded beforehand.

Two comprehension batteries appear in this paper and should not be confused. An earlier open-source audit uses a 41-probe bank over the same six domains, varying wording, paraphrase, calculator disclosure and answer order, and scoring format compliance separately from semantic correctness. The confirmatory frontier endpoint admission reported in §4.1 is a separate, frozen battery of 20 probes at three requested repetitions, for 60 retained answers per route. A paired behavioural diagnostic compares the canonical prompt against a deterministic four-row decision card that discloses the focal agent's immediate payoff, progress and private-risk consequence for each action profile, while revealing neither the opponent's move nor the stopping horizon.

Three further manipulations are reported in the supplementary material and described there in full: a persona framing assigned to each company's executive, in three families with a length-matched neutral placebo that separates the persona from the mere presence of an extra sentence; eight narrative skins that retell the same game as different stories; and a substitution of the opaque response codes P and Q for SAFE and UNSAFE. The last two are read three ways, as a paired first-round comparison at an identical state, as a *fixed-state replay* that asks for one action at a previously recorded state without letting the answer change later states, and as a *live trajectory* in which each answer does change the next state. A comprehension check gates all of them: if it fails, a behavioural difference shows prompt-conditioned output rather than informed play.

4 RESULTS

We report the validity gate first, because every strategic claim below depends on it, and then ask whether measured comprehension tracks behaviour at all. The remaining subsections answer the strategic question for the audited routes. Three- to five-player races, full model tables, predictive diagnostics and robustness checks are in the supplementary material.

Table 2: Semantic accuracy on the frontier endpoint admission, one row per audited route, ordered by overall accuracy. Each route contributes 20 probes at three requested repetitions, for 60 retained outputs. Reading the stage-payoff matrix was answered perfectly by every route, and rule recall by every route except GPT-5.4 nano (75.0%), so those two subtasks are omitted here. Expected payoff is recorded but does not gate admission.

Route	State recon.	State trans.	Term. score	Exp. pay (diag.)	Overall	Adm.
Gemini 3 Flash	93.3	100.0	100.0	50.0	93.3	yes
Claude Opus 5	93.3	100.0	100.0	33.3	91.7	yes
GPT-5.4	100.0	100.0	100.0	0.0	90.0	yes
GPT-5.5	80.0	100.0	100.0	50.0	90.0	yes
Claude Sonnet 5	100.0	100.0	80.0	0.0	85.0	yes
Gemini 3.1 Flash-Lite	73.3	100.0	80.0	16.7	80.0	no
GPT-5.4 mini	66.7	66.7	86.7	0.0	75.0	no
Gemini 3.5 Flash-Lite	60.0	50.0	60.0	0.0	65.0	no
GPT-5.4 nano	20.0	50.0	66.7	0.0	51.7	no

4.1 Validity gate: can agents follow the race?

We first ask whether a frontier endpoint can carry out the rules it receives. Nine routes were audited under one frozen probe bank: 20 atomic probes at three requested repetitions, so 60 retained outputs per route and 540 in total. Every planned output was retained and no parse failure occurred. Admission requires 80% overall accuracy together with 75% on state reconstruction and 75% on terminal scoring. The closed-form expected-payoff probes are recorded but do not gate admission, because the game asks an agent to choose an action, not to solve for an expected value.

Table 2 reports the result, and the striking feature is how unevenly comprehension is distributed. Two subtasks are solved by everyone: reading a value out of the fixed payoff matrix is answered perfectly by all nine routes, and stating the rules by eight of nine. Every difference between endpoints appears in the harder subtasks, where accuracy at rebuilding the current state from a short history ranges from 100% down to 20%. Five routes clear the gate; four do not, and GPT-5.4 nano reconstructs the state correctly in only 20% of probes while still returning a correctly formatted action every time it is asked to play.

Two points follow. No route reaches 75% on closed-form expected payoff and five score zero, so the arithmetic gap is general rather than a property of one weak endpoint. And a correctly formatted action carries no information about whether the agent tracked the state behind it: the parser was satisfied on all 540 outputs while state reconstruction varied across the full range. That is the concrete sense in which a valid action is not a valid measurement. Probe-level outputs, retained failure records, and unreachable routes are in the supplementary material.

4.2 Does measured comprehension track how play responds to risk?

The audit is only worth running if it tells us something about the behaviour it gates. We can now ask that directly, because all nine audited routes also completed the confirmatory two-player baseline under one frozen protocol: 30 races, 558 decisions, and zero parse failures each, with 10 races in every risk cell. We summarise each route's strategic sensitivity by its *risk response*, the UNSAFE-rate difference between the lowest and highest assigned private risk, resampled over races because the two seats of a race share its sampled horizon and setback draw.

Two routes also ran in the earlier confirmatory block, giving a temporal replicate of one frozen protocol. They agree closely: the largest per-cell difference is 1.1 percentage points for Gemini 3 Flash and 5.3 for Claude Sonnet 5, against risk responses of 38.7 and 57.0 points. Provider-side variation is thus small relative to the reported effect, which matters because no provider confirmed applying the requested seed.

The risk response spans almost the whole available range, from 7.0 percentage points for GPT-5.4 mini to 100.0 for Claude Opus 5 (Figure 2a), and it ranks with measured comprehension. Across the nine routes, the rank correlation between risk response and state-reconstruction accuracy is $\rho = 0.87$ (exact permutation $p = 0.004$, $n = 9$), rising to $\rho = 0.92$ ($p = 0.003$, $n = 8$) when Claude Opus 5 is set aside; that route plays UNSAFE on every decision at risk 0.1 and SAFE on every decision at both higher levels, a degenerate step policy whose within-cell variance is zero. We report the correlation with and without it rather than choosing one, because a single extreme route should not be able to carry a rank statistic computed over nine.

Three cautions belong with that number, and none is removable by more careful statistics. Nine commercial endpoints are not a sample from a population of models, so this is a descriptive association, not an estimated effect. The endpoints differ in far more than their probe scores, so nothing here separates comprehension from provider, scale, or training. And the gate does not partition the sample cleanly: Gemini 3.1 Flash-Lite misses admission on state reconstruction, 73.3% against the 75% threshold, yet responds to risk as strongly as three admitted routes (Figure 2b). The threshold is a declared entry condition, not a behavioural predictor.

What the pattern does establish is the practical point. The two routes whose play is flattest in risk, GPT-5.4 nano at 9.7 percentage points and GPT-5.4 mini at 7.0, score 20.0% and 66.7% on state reconstruction, both below the gate. Their UNSAFE rates near 60% would sit unremarkably in a cross-model table and are close to the human mean, yet they come from endpoints that largely cannot rebuild the state the risk applies to. An aggregate rate reported without a comprehension gate cannot tell that apart from a deliberate policy of moderate risk-taking. Across the four refused routes the risk response averages 19.4 points against 54.8 for the five admitted ones.

4.3 Strategic robustness under equivalent presentations

We next ask whether sampled behaviour is stable under manipulations that leave the underlying game mechanics unchanged.

Disclosed arithmetic. The canonical and decision-card conditions each completed 30 races and 558 decisions with zero parse failures; matched risk-by-repetition cells had identical realised horizons. Table 3 compares them, with intervals resampled over races rather than over decisions, because two decisions in one race share a state history and are not separate draws. Only 3.3% of paired first-round decisions changed, so most of the divergence appeared later in the histories, after feedback had accumulated. Supplying the current round's arithmetic shifts the sampled UNSAFE rate upward by about nine percentage points, but the two race-clustered intervals still touch at their edges, so we read this as a suggestive shift rather than an established effect, and certainly not as evidence of an internal world model.

Table 3: Disclosed-arithmetic diagnostic: canonical vs. decision-card conditions, 30 races and 558 decisions each, zero parse failures. Intervals are 95% percentile bootstrap intervals over the 30 races.

Condition	UNSAFE rate	95% CI	Mean payoff
Canonical	52.0%	48.9 to 55.4%	42.77
Decision card	60.8%	55.1 to 65.3%	42.21

Opaque symbolic mapping. The earlier local context-skin pilot confounded mapping with repetition parity and failed its comprehension gate, so it is excluded from the headline tables and retained only in the audit trail. A fully crossed per-endpoint rerun, counterbalancing narrative skin against the opaque P/Q response codes within repetition blocks, completed on Gemini 3 Flash: 120 races and 2,232 decisions. The matched Claude Sonnet 5 run stopped at 106 of 120 races on a provider quota refusal and is retained as a failure record rather than reported as a partial result. Because the design is crossed for one route only, we still make no general claim about symbol mapping or narrative skins; the completed cell is reported in the supplementary material.

Together, these diagnostics show that action sequences can shift under changes that a correct game model should ignore. This is why the comprehension check in §4.1 comes before behavioural interpretation.

4.4 Game theory versus prompted race behaviour

At the reference setting reported by Domingos and Han [14], selection strength $\beta = 2$ with mutation $\mu = \beta/Z$, the reconstructed evolutionary model predicts a switch rather than a gradient: UNSAFE play of 99.2%, 98.0%, and 1.9% at maximum risks 0.1, 0.6, and 0.9. The audited routes instead decline smoothly. In the earlier confirmatory block, Gemini 3 Flash plays UNSAFE 98.9%, 74.2%, and 59.1% and Claude Sonnet 5 89.2%, 46.2%, and 37.6%, each over 30 races and 558 decisions (Figure 3).

Read against that single reference setting, the prompted routes look like a failure to reproduce the theory. That reading does not survive a sensitivity check. We swept selection strength across $\beta \in [0.001, 10]$ crossed with three mutation rules, 90 stationary-distribution cells in total, each estimated from independent chains with the between-chain spread retained so a badly-mixed cell cannot be mistaken for a result. The sharp switch is a property of strong selection, not of the game: as β falls, the predicted high-risk UNSAFE rate rises off the floor and the risk response becomes graded. At $\beta = 0.01$ with $\mu = 0.05$ the same model predicts 87.3%, 63.9%, and 38.0%, and that profile is the closest cell in the entire sweep to both

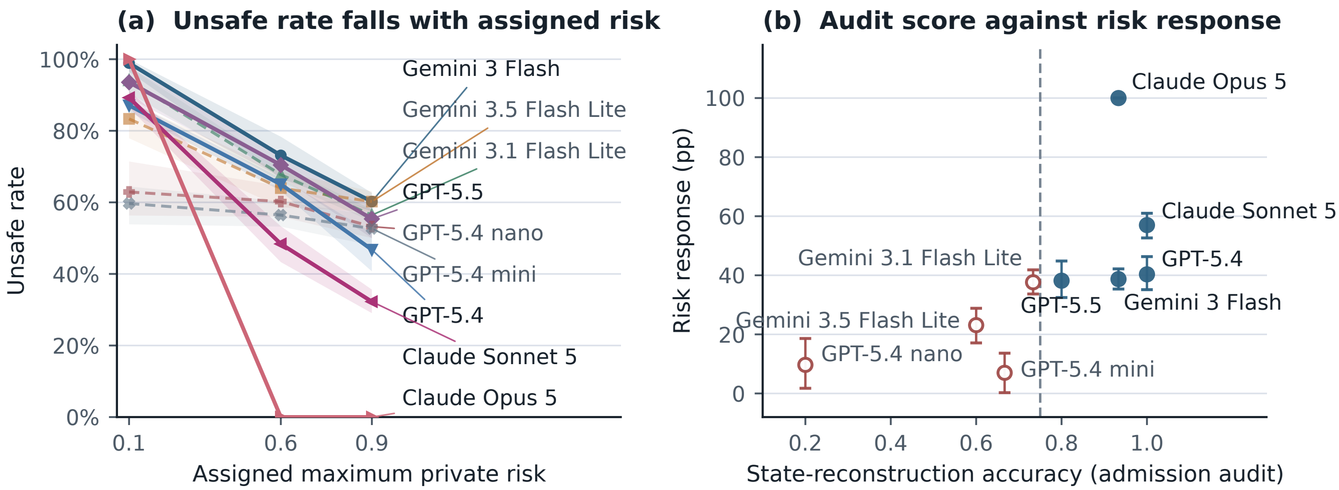

Figure 2: Measured comprehension against strategic sensitivity, over the nine audited routes, each of which completed the confirmatory baseline. (a) UNSAFE rate against assigned maximum private risk, with 95% race-clustered bootstrap bands; solid lines are admitted routes and dashed lines refused ones. (b) State-reconstruction accuracy from the admission audit against the risk response, the UNSAFE-rate difference between risk 0.1 and 0.9, with race-clustered intervals; filled points are admitted, hollow refused, and the dashed rule is the 75% admission threshold. Claude Opus 5 is a degenerate step policy with zero within-cell variance and is reported as a boundary case.

audited routes, within 10.3 percentage points of Claude Sonnet 5 and 15.1 of Gemini 3 Flash in root-mean-square error.

Two things follow. First, $\beta = 0.01$ and $\mu = 0.05$ are not parameters we chose to fit: they are the values the source study reports as its own best fit to human play, so the audited routes sit closer to the regime that describes the human participants than to the strong-selection regime usually quoted as the model's prediction. Second, the interpretation stays bounded. A stationary distribution describes strategy frequencies in a population under selection and mutation, while our measurements are prompted self-play decisions; the agreement in shape says the graded risk response is consistent with a near-neutral, high-exploration population, not that the routes are evolving. The sweep establishes something narrower and still useful: a claim that LLM agents contradict the evolutionary benchmark cannot rest on one setting of the selection strength, because the sign of the high-risk gap depends on it.

4.5 Two-player strategic diversity: humans and LLMs

We compare model trajectories with the human benchmark of Domingos and Han [14] directly on the literal first-five-round action-and-position histories, without first reducing them to hand-designed summary statistics. The comparison includes 420 trajectories from the seven tested models' two-player baseline (60 per model: GPT-5-nano, GPT-5.4-nano, Gemini-3-Flash, Gemini-3.1-Flash-Lite, Gemini-3.5-Flash-Lite, Claude Opus 5, and Claude Sonnet 5) and 340 complete human trajectories from the public de-identified dataset (one incomplete human record excluded); pooled $N = 760$. Six of these seven checkpoints now carry an endpoint-admission verdict from Table 2: Gemini 3 Flash, Claude Opus 5, and Claude Sonnet 5 passed the gate, while Gemini 3.1 Flash-Lite,

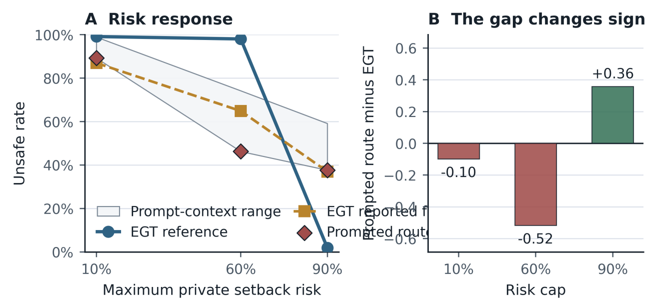

Figure 3: The prompted frontier does not reproduce the reconstructed evolutionary risk transition at the strong-selection reference setting. Panel A shows the stationary evolutionary reference, the reported fit, and the prompted route with its context range. Panel B makes the changing gap explicit: the prompted route is below the reference at medium risk but above it at high risk. The selection-strength sweep in §4.4 shows that this sign change is regime-dependent.

Gemini 3.5 Flash-Lite, and GPT-5.4 nano did not. GPT-5-nano is no longer offered on the audited identity and so cannot be gated at all; it is retained here as historical description. Each player is represented by 15 raw features: own action, opponent action, and own-minus-opponent progress gap entering each of rounds 1 to 5. These are the recorded states and actions themselves, not derived quantities such as unsafe rate or retaliation. The archetype projection below uses 341 participants rather than 340, because its five descriptor summaries can be computed for one further participant whose complete five-round sequence is unavailable.

Mean UNSAFE play over rounds 1 to 5 is lowest for GPT-5-nano (17%), followed by Claude Sonnet 5 (22%), Claude Opus 5 (33%), GPT-5.4-nano (54%), and humans (56%); the Gemini models range from 73% (Gemini-3-Flash) to 83% (Gemini-3.1-Flash-Lite). No single aggregate rate characterises the tested models, and closeness to the human mean is, on its own, uninformative about anything beyond that one number.

A complementary view projects each trajectory onto the nearest of four behavioural archetypes fitted to the human sample alone, using five descriptors: overall UNSAFE rate, opponent reciprocity, position sensitivity, own-action persistence, and first-round UNSAFE play. The archetype signatures, the per-population coverage shares, and a sweep of the cluster count from two to six with a bootstrap stability curve are all in the supplementary material. Two facts from it matter here. The human sample occupies every archetype at every cluster count tested, while each checkpoint except GPT-5.4 nano is more concentrated, so that conclusion does not depend on the choice of four. And the archetypes are exploratory descriptions rather than recovered types: the four-group solution is the best separated on both silhouette and Davies–Bouldin over the sweep, but its bootstrap stability is only moderate.

A t-SNE embedding of the same pooled 15-dimensional space, reported in the supplementary material, makes the distinction between behavioural *level* and behavioural *diversity* visible: each checkpoint concentrates in a narrower region than the human sample, which overlaps several model-specific regions. We do not rest any conclusion on that embedding. Recolouring the identical coordinates by outcome rather than by identity separates it almost entirely by UNSAFE rate, so the apparent separation between populations is largely a separation between action rates. The diversity comparison is therefore made directly on the action sequences, with no embedding and no clustering step.

How much diversity, measured directly. We compare the exact paired five-round action sequence of each player, a ten-bit string, within each risk condition; because the progress increments are fixed, that string is the whole trajectory. Two statistics are reported, because either alone can mislead. Hill q=1, the exponentiated Shannon entropy, is the effective number of distinct sequences a population produces, and is blind to how different they are; the mean pairwise Hamming distance measures how far apart they are, and is blind to how many there are. Every cell is rarefied to 20 trajectories so the human sample, five to seven times larger, cannot look more diverse merely for being bigger, and every interval is a cluster bootstrap over races, or over participants for the humans, since both seats of a race share its sampled horizon and setback draw.

Figure 4 gives the result, and it is unusually clean. Humans produce close to the maximum possible variety, an effective 18.9, 19.7, and 19.3 distinct sequences out of 20 at risk 0.1, 0.6, and 0.9, at mean pairwise distances of 0.47, 0.50, and 0.50. Six of the seven checkpoints fall entirely below the human interval on Hill q=1 in all three conditions. The compression at the extreme is severe rather than marginal: Claude Opus 5 produces a single sequence, repeated by all 20 of its trajectories, at every risk level, and Gemini 3 Flash and Gemini 3.1 Flash-Lite do the same at risk 0.1. A population of twenty agents collapsing to one policy is not a narrower distribution of strategies; it is the absence of one.

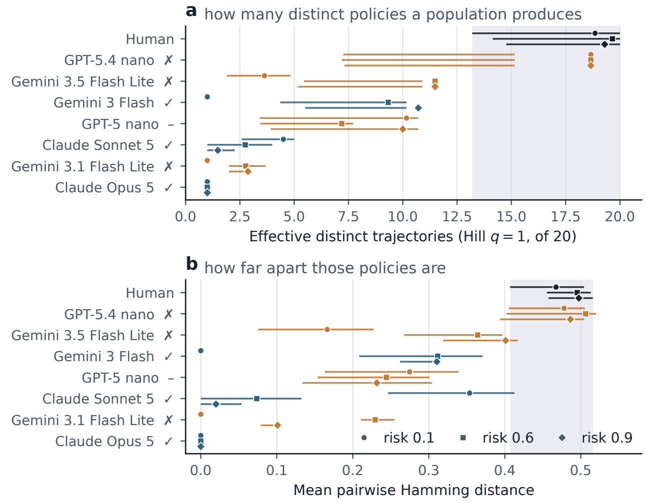

Figure 4: Within-population diversity of the exact paired five-round action sequence, by risk condition. Panel A counts distinct policies, panel B measures how far apart they are. Every cell is rarefied to 20 trajectories; bars are 95% cluster-bootstrap intervals over races, or over participants for the human sample. The shaded band spans the human interval. A tick marks a checkpoint that passed the endpoint-admission gate of Table 2, a cross one that failed, and a dash GPT-5-nano, which is no longer reachable and so has no verdict. Every admitted checkpoint sits below the human band; the one checkpoint that reaches it, GPT-5.4 nano, failed the gate.

One checkpoint is the exception, and it is the most informative row in the figure. GPT-5.4 nano reaches an effective 18.7 distinct sequences at every risk level, with intervals that overlap the human reference on both statistics, and its mean pairwise distances of 0.48, 0.51, and 0.49 are indistinguishable from the human values. On this measure it is not less diverse than people. It is also the checkpoint that fails the comprehension gate by the widest margin in Table 2, reconstructing the state correctly in 20% of probes and scoring 51.7% overall. Among the checkpoints that pass the gate, every one is strictly less diverse than the human sample at every risk level, with no exception. The audit is therefore not a formality placed in front of the interesting result; it is what separates a population whose variety reflects strategy from one whose variety is consistent with not tracking the game. Reported without the gate, this single row would read as the headline finding that one small model matches human strategic diversity.

The choice of four human archetypes above does not drive this. Sweeping k from two to six, the four-group solution is the best separated on both silhouette (0.32, against 0.28 at k=2 and 0.24 at k=5) and Davies–Bouldin (1.22, the minimum over the sweep), though bootstrap stability is only moderate (mean adjusted Rand index 0.63, 95% interval 0.39 to 0.98), which is why we treat the groups as exploratory descriptions rather than recovered types. The coverage conclusion holds at every k from two to six: the human sample occupies all k archetypes while every checkpoint except

GPT-5.4 nano is more concentrated. The sweep and its stability curve are in the supplementary material.

A shallow classifier trained on the same raw sequences, with folds grouped by race, recovers the population that produced a trajectory well above a grouped permutation null (balanced accuracy 42.3% ± 4.3 against a null mean of 12.4%, $p < 0.001$). Population identity is thus detectable, but the groups overlap, and this is evidence of observable trajectory differences rather than of a recoverable latent strategy. The diagnostic and its source hashes are in the supplementary material.

This heterogeneity is not merely a visual impression. Fitting three nested logistic models to the pooled neutral-baseline pilot (risk-only, LLM-identity-only, and LLM-identity-by-risk interaction) shows that allowing the risk-response profile to vary by LLM improves fit substantially beyond LLM-specific intercepts alone,

$$\chi^2(10) = 354.7, \qquad p < 10^{-3},$$

with several risk-by-LLM cells approaching complete separation. A convergence warning in the interaction model means that the displayed likelihood-ratio statistic is approximate, so we report the conservative threshold rather than false precision. The fit covers the five-model neutral baseline only; the two Claude models are not in it. Neither caveat changes the conclusion: no single risk-response curve represents all five, so cross-model differences are differences in *shape*, not only in level.

The same picture holds for the raw player-level UNSAFE rates, plotted per risk condition in the supplementary material: human rates spread across nearly the whole available range at every risk level, while each checkpoint occupies a narrow band of its own. Similarity between a checkpoint's aggregate UNSAFE rate and the human mean therefore does not imply that it reproduces the human distribution.

4.6 Observable drivers of UNSAFE play

We also ask which pre-decision variables predict UNSAFE play: the player's own previous action, the opponent's previous action, the progress gap, the assigned private risk, and the round number. These are predictive associations, not causal effects. Fits are cross-validated with folds grouped by race, and the full random-forest and SHAP analysis is in the supplementary material.

One marginal association does agree across populations: human winners play UNSAFE 16.0 percentage points more often than human losers, against an approximately 20-point pooled difference among the checkpoints. The dynamic structure behind it does not agree. The human data are organised mainly by the opponent's previous action, which accounts for 56% of the fitted model's total SHAP magnitude. The tested checkpoints do not share one pattern. Progress gap is largest for GPT-5-nano (41%), the assigned risk condition is largest for two Gemini checkpoints (33% and 37%), and opponent history is largest for Claude Sonnet 5 (48%). GPT-5.4-nano has no single strong predictor. These results support a game-theoretic point: two populations can have similar UNSAFE rates while following different response rules. Several checkpoint estimates are also weak, reversed in sign, or not reliably estimable because their action distributions sit at the floor or the ceiling. The supplementary material reports model fit, confounding, and those floor and ceiling limits.

5 DISCUSSION AND CONCLUSION

The lesson is not that an audited route "understands the game". The evidence supports something narrower and more useful. Admitted routes recall the rules, rebuild the tested states and apply terminal eligibility, yet none reliably computes a closed-form expected payoff, and the routes that fail the gate still return a perfectly formatted action every time. A study that reports action rates without per-route comprehension accuracy and parser health cannot tell those situations apart, and they license opposite conclusions.

One contrast is worth keeping in view, because it is so often conflated. The human participants' incentivised pre-game risk score has almost no relation to their later UNSAFE play ($r = -0.015$, $p = 0.79$, $n = 341$) and does not even separate the behavioural archetypes, whereas sweeping a prompted risk persona from its lowest to its highest level moves UNSAFE play by 50.5 to 98.3 percentage points. A measured preference describes a person before play; a persona label rewrites the decision context during it. A persona is a strong policy prompt, not evidence of a stable risk preference, and the curves are in the supplementary material.

The human experiment is an external behavioural reference, not a target an agent should reproduce. Where a route matches a human association, the two populations behave similarly on that one measured aspect of this task; it does not follow that they share a mechanism, and a mismatch does not invalidate the human result. Every comparison here is scoped to its tested route, prompt version, decoding contract, provider path and run period.

Three limitations bound the claims directly. The comprehension-to-behaviour association covers nine endpoints that differ in far more than their probe scores, so it is descriptive and cannot separate comprehension from provider, scale or training. The representation-robustness design is fully crossed for one route only, since the matched second route stopped on a provider quota refusal, so no general claim about symbol mapping follows. And the evolutionary comparison is a repo-native reconstruction audited against pinned EGTtools source rather than a bitwise reproduction of undisclosed author code, so its role is to bound the regime in which the benchmark is read, not to fit the agents.

Four further limits are worth stating. Fixed-state replay and live play measure different things, one decision at a saved state against dependent decisions along a changing trajectory, so their effect sizes are not interchangeable. The multiplayer pilots are small, have no matched two-player condition, and vary the payoff rule, the opponent count and the prompt length together, so no pure group-size effect is identifiable. Seed forwarding is provider-specific: one route's SDK stripped the requested seed and another forwarded it unconfirmed. And this game models a private speed-versus-safety trade-off and nothing else, with no collective harm, regulation, communication, or uncertainty about real capabilities.

Strategic safety behaviour in these races is not a fixed property of an LLM. Across nine audited routes, rule recall does not ensure state tracking and an equivalent presentation can change later play. The audit is what makes the behavioural differences readable: every route that passes it is less diverse than the human participants and responds strongly to the assigned risk, while the routes that fail it include both the one that most resembles human diversity and the ones whose play is nearly flat in risk.

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